26F Stat TA Session W4

Generate Random Numbers

- After 3 weeks of basic data manipulation and plot-drawing (data visualization), you may expect that we are still going to draw figures today. But Hey, this is a probability and statistics course!
- We (should) have learned in the lecture the concept of **random variables** and **distributions**:

> 💡 **Random Variable:**
>
> A function that assigns a value to each outcome in the sample space
>
> **Distribution:**
>
> A description of how probabilities are assigned to the values of a random variable

- **Example:** Roll a Die
    - Let X be the r.v. that equals the number of dots ($\in \{1, 2, 3, 4, 5, 6\}$) on the upper face
    - Each value has probability $1/6$

Random Seed

- Now we are going to draw numbers from certain distributions in `R` , but how can we allow others to reproduce our results if we are drawing randomly?
- A seed allows us to do so by specifying a sampling function: `runif()` , `rnorm()` , `sample()`

```
set.seed(20261001) rnorm(1, mean = 0, sd = 1) #output [1] -0.6833539 # the output should always be the same!
```

Uniform Random Samples `runif()`

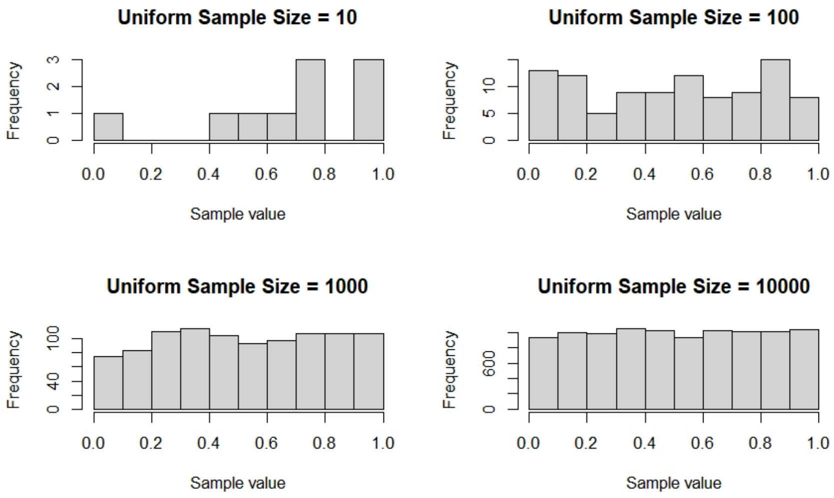

Figure: Random Samples drawn from Uniform[0,1] in different sizes

```
rdnum <- runif(10) # default: min = 0, max = 1 rdnum_2 <- runif(n = 10, min = 0, max = 100) # output print(rdnum) [1] 0.60948729 0.12423338 0.84172306 0.87140030 0.21421948 [6] 0.63042043 0.94701880 0.83829748 0.20671720 0.06368964 print(rdnum_2) [1] 80.18057 92.06480 94.93640 92.88272 96.57980 [6] 29.93912 72.90744 39.00587 16.61714 22.60201
```

Coin Toss

❓ **Q.** How can we simulate a coin toss game in `R` ?

- What do we have in coin toss?
  - A fair coin with equal probability of being head or tail
- A uniform distribution on $[0, 1]$ can be divided into equal halves!

```
head <- rdnum > 0.5 class(head) # "logical" # output [1] TRUE FALSE TRUE TRUE FALSE [6] TRUE TRUE TRUE FALSE FALSE
```

- `head` gives `TRUE` or `FALSE` . If we want to do calculations, we have to convert them into numbers!

```
# head gives TRUE or FALSE # to do calculations, we want them to be numbers head_numeric <- as.numeric(head) # output [1] 1 0 1 1 0 1 1 1 0 0
```

- `as.numeric()` convert things to numbers. Similarly, we have `as.logical()` , `as.integer()` , `as.character()`
- With numeric values, we can calculate the mean and it's more useful for data analysis!

```
mean(head_numeric) # 0.6 rdnum_3 <- runif(10000) head_numeric_2 <- as.numeric(rdnum_3 >
0.5) mean(head_numeric_2) # 0.4995 -> closer to 0.5! why?
```

Operators: `$` , `[]` , `[[]]`

- For data cleaning, we may find undesirable data types. But now we can easily convert them into the data type we want!

```
data_example <- data.frame(name = c("Alice", "Bob", "Cindy"), score = c("80", "60",
"100") ) mean(data_example$score) # NA! mean(as.numeric(data_example$score)) # 80
```

- In above example, we use `$` to extract the column we want
- We can also use `[]` or `[[]]`

▶ **(Optional) Q. But what's the difference?**

Cumulative Sum of Sequences `cumsum()`

- With an indicator of being head or not, we can count the cumulative number of heads using `cumsum()`

```
cum_head <- cumsum(head_numeric) # output [1] 1 1 2 3 3 4 5 6 6 6 # we have 6 heads in
10 tosses
```

Exercises

W4 Exercises